MIL-B-45326B(AT)
1 October 1976
SUPERSEDING
MIL-B-45326A(MO)
18 November 1964

MILITARY SPECIFICATION

BOOT, DUST AND MOISTURE SEAL: HYDRAULIC BRAKE CYLINDER, SYNTHETIC RUBBER

This specification is approved by Army Tank-Automotive Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers brake boots used in covering the open ends of automotive brake cylinders employing hydraulic brake fluid of the nonpetroleum base type.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Federal
VV-H-910 - Hydraulic Fluid, Non-Petroleum Base, Automotive, Brake.

STANDARDS
Federal
Fed. Test Method Std. No. 601 - Rubber: Sampling and Testing.

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items.
MIL-STD-417 - Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests).

FSC-2530

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (insert name and address of the preparing activity) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Initial production sample. Boots furnished under this specification shall be products which have been represented by initial production samples that have been tested and have passed the initial production tests specified herein (see 4.2). Approval of the initial production sample by the activity concerned shall not relieve the contractor of his obligation to supply items as specified herein. Any changes or deviations of the production units from the initial production samples shall be subject to the approval of the contracting officer.

3.2 Material. Material shall be a synthetic rubber compound conforming to RS 515B or RS 615B of MIL-STD-417 with the following exceptions and additions:

3.2.1 Durometer hardness. Durometer hardness shall be determined for each boot in each test sample before further testing and shall be determined again after testing as specified in 4.5.2 and 4.5.5. When boots are acceptance or control tested, their durometer shall be equal to their qualifying value within plus or minus 5 points, providing the durometer hardness is within the limits of 45 to 65 points.

3.2.2 Volume change. The volume of the test specimen shall neither increase by more than 20 percent nor decrease by more than 5 percent when tested as specified in 4.5.2. The durometer hardness shall not increase by more than 5 points nor decrease by more than 10 points after testing.

3.2.3 Heat stroking. Test specimen shall be free from flex cracks and shall fit snugly around cylinder and push rod after being tested as specified in 4.5.3.

3.2.4 Low temperature flexibility. Test specimens shall not crack when tested as specified in 4.5.4.

3.2.5 Heat resistance. When tested as specified in 4.5.5, durometer hardness shall be not greater than 15 points higher or not more than 5 points lower than the pretest durometer hardness. The rubber, after cooling, shall not be tacky and there shall be no cracking when folded.

3.2.6 Ozone resistance. The specimen shall exhibit no evidence of cracking or rupture when examined under 7-power magnification after being tested as specified in 4.5.6.

3.3 Design and construction. Boots shall be of molded construction and shall conform to the dimensions specified on the applicable drawings (see 6.2).

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3.4 Marking. The identification mark of the manufacturer, the capital letters "OZ", and other marking as specified on applicable drawings or by the procuring activity (see 6.2), shall be molded into each boot.

3.5 Workmanship. Workmanship shall be such that the defects specified in 4.3.2.2 shall not be present.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Initial production test. An initial production sample of ten boots of each size and design shall be furnished for initial production inspection. Samples shall be the first units produced under the contract and representative of the units proposed to be furnished under the contract. All inspections shall be done by the manufacturer under Government surveillance and shall consist of examination for defects and testing in accordance with Table I.

4.2.1 Failure. Failure of an initial production sample to pass any of the examinations or tests specified herein may be cause for the Government's refusal to witness additional examinations or tests unitl the faults revealed have been corrected.

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Lot formation. Unless otherwise specified (see 6.2), a lot shall consist of all boots of one size and design, from an identifiable production period, produced by one manufacturer and submitted at one time for acceptance.

4.3.1.2 Sampling for quality conformance examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.3.1.3 Sampling for quality conformance testing. Samples for quality conformance testing shall be selected in accordance with inspection level S3 of MIL-STD-105.

4.3.2 Examination inspection for boots.

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4.3.2.1 Acceptable quality levels. Each sample selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL), on the basis of percent defective:

Classification	AQL
Major defects	2.5
Minor defects	4.0

4.3.2.2 Classification of defects. For examination purposes, defects shall be determined and classified as specified in MIL-STD-407.

4.3.3 Testing inspection for boots. Each sample selected in accordance with 4.3.1.3 shall be tested as specified in Table I to determine conformance to 3.2.1.

4.4 Control tests.

4.4.1 Sampling and testing for boots. Control test samples shall be selected at the rate of one sample (consisting of 10 boots) of each 1000 boots produced, except that not less than one nor more than two samples shall be selected in any 30 day period. Boots shall be examined for the defects specified in 4.3.2.2, and subsequently tested as specified in Table I.

Table I. Table of tests

Sample	Test	Initial Production	Testing Inspection	Control
All	4.5.1 - Durometer hardness	X	X	X
1, 2	4.5.2 - Volume change	X		
3, 4	4.5.3 - Heat stroking	X		
5, 6	4.5.4 - Low temperature flexibility	X		X
7, 8	4.5.5 - Heat resistance	X		
9, 10	4.5.6 - Ozone resistance	X		X

4.5 Test procedures

4.5.1 Durometer hardness. To determine conformance to 3.2.1, sample boots shall be tested for durometer hardness in accordance with Method 3021 of Fed. Test Method Std. No. 601.

4.5.2 Volume change. Volume change shall be determined in accordance with Method 6211 of Fed. Test Method Std. No. 601. Test fluid shall conform to VV-H-910. Test fluid temperature shall be 250° ± 5°F. Time of immersion

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shall be 70 ± 1/4 hours. Two specimens shall be tested. Each specimen shall be cut from a separate boot and shall be approximately 1 square inch in area. At completion of test, specimens shall be examined to determine conformance to 3.2.2.

4.5.3 Heat stroking.

4.5.3.1 Apparatus. Stroking devices shall be provided, consisting of two master cylinder assemblies and two wheel cylinder assemblies or equivalent, on which test boots can be mounted, with appropriate mechanism to permit cycling as specified herein. An oven, of the type described in Method 7221 of Fed. Test Method Std. No. 601, shall be provided.

4.5.3.2 Method. Two boots of one type shall be mounted on the stroking device. The push rods shall then be cycled at the rate of 1000 ± 100 strokes per hour for 120,000 cycles. Stroke length shall be 80 percent of the maximum stroke for master cylinders and 3/16 inch for wheel cylinders. Ambient air temperature shall be 158° ± 2°F. No oil shall contact the boots during stroking. After the specified cycles have been completed, the boot and cylinder assemblies shall be allowed to cool to room temperature. The boots shall be examined for flex cracks extending through the wall thickness and for fit around cylinder and push rod to determine conformance to 3.2.3.

4.5.4 Low temperature flexibility.

4.5.4.1 Apparatus. Test fixtures shall be provided, consisting of two master cylinder assemblies and two wheel cylinder assemblies or equivalent, on which test boots can be mounted and tested as specified. When boots are designed to slide on the push rods, collars or similar locking devices shall be furnished to prevent relative motion between boot and push rod during test. A cold chamber shall be provided which will contain the test fixture, permit observation and actuation of the fixture mechanism, and maintain a dry air temperature of minus 65° ± 3°F.

4.5.4.2 Method. Two boots of one type shall be mounted on the fixture, placed in the cold chamber, and held at a temperature of minus 65° ± 3°F. for not less than 70 hours. The boots shall be so mounted that relative motion between each boot and push rod will not occur. The push rods shall then be actuated through the maximum stroke, for which the cylinder was designed, for a total of 10 cycles. The boots shall then be removed from the cold chamber and examined for cracks or other damage, to determine conformance to 3.2.4.

4.5.5 Heat resistance.

4.5.5.1 Apparatus. A circulating-air oven as described in Method 7221 of Fed. Test Method Std. No. 601, with provisions for suspending boots within the oven, shall be provided.

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4.5.5.2 Method. Two boots of one type shall be tested. One boot shall be suspended in the oven for 70 ± 1/4 hours at 212° ± 5°F. and one boot shall be suspended in the oven for 24 ± 1/4 hours at 250° ± 5°F. To determine conformance to 3.2.5, the durometer hardness of each boot shall be determined after oven aging and each boot shall be folded and examined for cracks and for tackiness.

4.5.6 Ozone resistance.

4.5.6.1 Sample size. Two boots shall be tested.

4.5.6.2 Method. Samples shall be assembled on a hydraulic brake cylinder or equivalent which will provide a 12 to 15 percent stretch in both the boot-bead and push-rod mounting areas. Samples shall then be conditioned in an ozone-free atmosphere at room temperature for a maximum of 45 minutes. Samples shall then be placed in an ozone chamber with the ozone concentration maintained at 50 ± 5 parts per hundred million parts of air by volume, at a temperature of 100° ± 2°F, for a period of 168 ± 1/2 hours. The boots shall be examined daily for cracking and rupture under 7-power magnification. At completion of test, boots shall be again examined under same magnification to determine conformance to 3.2.6.

4.6 Inspection of preparation for delivery.

4.6.1 Materials and processes. The Government inspector shall at unscheduled intervals inspect all materials and processes involved in the preparation for delivery to determine conformance to the requirements of section 5. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with the packaging data sheet, as furnished at the time of invitation for bids, for the level of preservation and packaging specified (see 6.2(f)).

6. NOTES

6.1 Intended use. Brake boots covered by this specification are intended for use as protective devices against dirt and moisture on wheel and master cylinders in hydraulic brake systems of automobiles, trucks, and other wheeled vehicles employing hydraulic brake fluid of the nonpetroleum base type.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of applicable drawings (see 3.3).
(c) Identification marking required (see 3.4).
(d) When contractor is not responsible for the performance of inspection requirements (see 4.1).
(e) When lot shall be other than specified (see 4.3.1.1).
(f) Selection of applicable level of preservation and packaging and level of packing (see 5.1).

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS. This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-B-45326B(AT)

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

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